

1	PIEZOELECTRIC MOLECULAR CRYSTAL LITERATURE DATABASE
2	Complete 8-Tier Crystal-Type File Hierarchy
3	=====
4	Scope: ONLY Molecular Crystals with Piezoelectric Properties
5	Total target crystals: 48 across 8 tiers
6	
7	PiezoResearch/
8	└─ 01_PDFs/
9	
10	├─ T0_review_articles/ ← Read first; cross-tier context
11	├─ Guerin2019_organic_piezo_milestones_NPGAsia.pdf
12	├─ Author_Year_piezo_biomaterials_chemrev.pdf
13	└─ Vishnoi2025_CrystalDFT_highthroughput.pdf
14	
15	
16	├─ T1_amino_acid_crystals/ ← TIER 1: MUST-HAVE (12 crystals)
17	Highest literature coverage; start here
18	├─ glycine_polymorphs/
19	├─ gamma_glycine/ ← d33 ~10 pC/N; trigonal P31/P32
20	├─ Author_Year_gammaGlycine_d33_measurement.pdf
21	└─ Author_Year_gammaGlycine_PFM_SHG.pdf
22	├─ beta_glycine/ ← Metastable; monoclinic P21
23	└─ Guerin2018_betaGlycine_NatMater.pdf
24	└─ alpha_glycine_doped/ ← Centrosymmetric host; piezo via doping only
25	└─ DishonBenAmi2022_alphaGlycine_doped.pdf ← HAVE
26	├─ L_alanine/ ← P212121; well-studied d14, d25, d36
27	└─ Author_Year_LAlanine_piezo_spacegroup.pdf
28	├─ L_valine/
29	└─ Author_Year_LValine_crystal_piezo.pdf
30	├─ L_leucine/
31	└─ Author_Year_LLeucine_piezo_PFM.pdf
32	├─ L_isoleucine/
33	└─ Author_Year_LIsoleucine_crystal_piezo.pdf
34	├─ L_lysine/ ← DFT-predicted high piezo; see CrystalDFT
35	└─ Author_Year_LLysine_piezo_DFT.pdf
36	├─ L_arginine/
37	└─ Author_Year_LArginine_crystal_piezo.pdf
38	├─ L_histidine/
39	└─ Author_Year_LHistidine_crystal_piezo.pdf
40	├─ L_threonine/ ← Also used as dopant in alpha-glycine
41	└─ Author_Year_LThreonine_crystal_piezo.pdf
42	├─ L_serine/ ← Also used as dopant in alpha-glycine
43	└─ Author_Year_LSerine_crystal_piezo.pdf
44	└─ hydroxy_L_proline/ ← Collagen subunit; unique H-bond structure
45	└─ Author_Year_HydroxyProline_piezo.pdf
46	
47	
48	├─ T2_peptide_crystals/ ← TIER 2: HIGH d33 VALUES (7 crystals)
49	Often higher piezo than amino acids
50	├─ dipeptides/
51	├─ diphenylalanine_FF/ ← Benchmark dipeptide; d33 up to 35 pC/N

52					└─ Author_Year_FF_nanotube_selfassembly_piezo.pdf
53					└─ Author_Year_FF_d33_PFM_measurement.pdf
54					└─ Yuan2023_FW_FF_coassembly_d33_35pmV.pdf
55				└─ cyclo_Phe_Phe/	← Cyclic dipeptide; enhanced non-centrosymmetry
56				└─ Author_Year_cycloPP_crystal_piezo.pdf	
57				└─ oligopeptides/	
58				└─ Gly_Phe_Gly/	
59				└─ Author_Year_GFG_tripeptide_piezo.pdf	
60				└─ Phe_Phe_Gly/	
61				└─ Author_Year_FFG_tripeptide_piezo.pdf	
62				└─ Ala_Gly_Gly/	← Collagen-related AHG subunit
63				└─ Author_Year_AGG_collagen_piezo.pdf	
64				└─ self_assembled_peptide_crystals/	← Fmoc-FF, hydrogel-derived crystals
65				└─ Author_Year_selfassembly_peptide_piezo_review.pdf	
66					
67					
68				└─ T3_organic_ferroelectric/	← TIER 3: STRONGEST PIEZO (7 crystals)
69					Switchable polarisation; often >> amino acids
70				└─ organic_acids/	
71				└─ croconic_acid/	← Proton-tautomer ferroelectric; huge P
72				└─ Author_Year_croconicAcid_ferroelectric_piezo.pdf	
73				└─ malonic_acid/	
74				└─ Author_Year_malonicAcid_piezo_crystal.pdf	
75				└─ succinic_acid/	
76				└─ Author_Year_succinicAcid_piezo.pdf	
77				└─ tartaric_acid/	← Rochelle salt precursor; historic piezo
78				└─ Author_Year_tartaricAcid_piezo_classic.pdf	
79				└─ ammonium_molecular_ferroelectrics/	← DIPA halides; record-setting organic piezo
80				└─ DIPAB_diisopropylammonium_bromide/	← d33 ~ 23 pC/N; near BaTiO3
81				└─ Author_Year_DIPAB_ferroelectric_Science.pdf	
82				└─ DIPAC_diisopropylammonium_chloride/	
83				└─ Author_Year_DIPAC_piezo_ferroelectric.pdf	
84				└─ DIPAI_diisopropylammonium_iodide/	
85				└─ Author_Year_DIPAI_piezo_crystal.pdf	
86					
87					
88				└─ T4_hydrogen_bonded/	← TIER 4: H-BOND DRIVEN (4 examples)
89					Polarisation from proton ordering / H-bond networks
90				└─ CBDC_cyclobutene_dicarboxylic_acid/	← Proton-ordered; switchable H-bond piezo
91				└─ Author_Year_CBDC_Hbond_piezo.pdf	
92				└─ PhMDA_phenylmalondialdehyde/	← Strong intramolecular H-bond; ferroelectric
93				└─ Author_Year_PhMDA_ferroelectric_piezo.pdf	
94				└─ croconic_acid_derivatives/	← H-tautomer family
95				└─ Author_Year_croconicAcid_derivative_piezo.pdf	
96				└─ Hbond_cocrystals/	← Multi-component H-bond engineered crystals
97				└─ Author_Year_Hbond_cocrystal_piezo.pdf	
98					
99					
100				└─ T5_biomolecular/	← TIER 5: BIOLOGICAL MOLECULES (8 crystals)
101					Nucleobases + metabolite crystals
102				└─ nucleobases/	← DNA/RNA building blocks in crystal form

103				└─ adenine/	
104				└─ Author_Year_adenine_crystal_piezo.pdf	
105				└─ guanine/	
106				└─ Author_Year_guanine_crystal_piezo.pdf	
107				└─ cytosine/	
108				└─ Author_Year_cytosine_crystal_piezo.pdf	
109				└─ uracil/	
110				└─ Author_Year_uracil_crystal_piezo.pdf	
111				└─ thymine/	
112				└─ Author_Year_thymine_crystal_piezo.pdf	
113				└─ metabolite_crystals/	
114				└─ urea/	← Classic non-centrosymmetric; NLO + piezo
115				└─ Author_Year_urea_crystal_piezo_NLO.pdf	
116				└─ creatinine/	
117				└─ Author_Year_creatinine_crystal_piezo.pdf	
118				└─ uric_acid/	
119				└─ Author_Year_uricAcid_crystal_piezo.pdf	
120					
121					
122				└─ T6_sugar_based/	← TIER 6: SUGAR CRYSTALS (4 crystals)
123					Chiral → non-centrosymmetric; biocompatible
124				└─ sucrose/	← P21 space group; known piezo
125				└─ Author_Year_sucrose_crystal_piezo.pdf	
126				└─ glucose/	
127				└─ Author_Year_glucose_crystal_piezo.pdf	
128				└─ lactose/	
129				└─ Author_Year_lactose_crystal_piezo.pdf	
130				└─ mannitol/	← Pharmaceutical; chiral; multiple polymorphs
131				└─ Author_Year_mannitol_piezo_crystal.pdf	
132					
133					
134				└─ T7_amino_acid_salt/	← TIER 7: AA SALT CRYSTALS (6+ salts)
135					Huge literature; non-centrosymmetric via protonatio
n					
136				└─ glycine_salts/	
137				└─ glycine_lithium_nitrate_GLN/	← Most studied; GLN benchmark salt
138				└─ Author_Year_GLN_piezo_NLO.pdf	
139				└─ glycine_phosphate/	
140				└─ Author_Year_glycinePhosphate_piezo.pdf	
141				└─ glycine_sulfate/	
142				└─ Author_Year_glycineSulfate_crystal_piezo.pdf	
143				└─ arginine_salts/	
144				└─ L_arginine_phosphate_LAP/	← Used in laser NLO; well-characterised
145				└─ Author_Year_LAP_piezo_NLO_laser.pdf	
146				└─ histidine_salts/	
147				└─ L_histidine_hydrochloride/	
148				└─ Author_Year_LHisHCl_crystal_piezo.pdf	
149				└─ alanine_salts/	
150				└─ L_alanine_acetate/	
151				└─ Author_Year_LAlanineAcetate_piezo.pdf	
152					

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153 |
154 |─ T8_charge_transfer/          ← TIER 8: ORGANIC CT CRYSTALS (3+ crystals)
155 | |
156 | |─ TTF_CA_tetrathiafulvalene_chloranil/ ← Neutral-ionic phase transition; classic CT
157 | | |─ Author_Year_TTFCA_ferroelectric_piezo.pdf
158 | | |─ Author_Year_TTFCA_neutralionic_transition.pdf
159 | |─ TTF_derivatives/          ← TTF-BA, TTF-BrCA, modified donors
160 | | |─ Author_Year_TTF_derivative_ferroelectric.pdf
161 | |─ chloranil_complexes/      ← Acceptor-side variants
162 | | |─ Author_Year_chloranil_complex_molferro.pdf
163 |
164 |─ _to_process/                ← INBOX: newly downloaded PDFs
165 | |─ (drop all new PDFs here before extracting with Claude)
166
167
168 PDF NAMING CONVENTION
169 ───────────────────
170 Format:  Author_Year_CrystalName_JournalOrKeyword.pdf
171
172 T1 examples: Guerin2018_betaGlycine_NatMater.pdf
173 T2 examples: Yuan2023_FF_dipeptide_d33_35pmV.pdf
174 T3 examples: Horiuchi2010_DIPAB_ferroelectric_Science.pdf
175 T4 examples: Author_Year_CBDC_Hbond_piezo.pdf
176 T5 examples: Author_Year_urea_crystal_NLO_piezo.pdf
177 T6 examples: Author_Year_sucrose_P21_piezo.pdf
178 T7 examples: Author_Year_GLN_glycineLiNitrate_piezo.pdf
179 T8 examples: Author_Year_TTFCA_chargetransfer_ferro.pdf
180
181
182 DATABASE EXTRA FIELDS PER TIER
183 ───────────────────
184 All tiers share: title, authors, DOI, year, journal, space_group,
185                  d_coefficients (pC/N or pm/V), measurement_method,
186                  temperature_range, crystal_system, polar_axis
187
188 Tier-specific additions:
189   T1 Amino acid → polymorph (α/β/γ), chirality (L/D/DL), dopant, doping_mol_pct
190   T2 Peptide   → sequence, length (di/tri/oligo), assembly_type, protection_group
191   T3 Ferroelec. → Ps (spontaneous polarisation μC/cm²), Ec (coercive field), Tc (Curie T)
192   T4 H-bond     → H-bond_type (inter/intra), proton_order (yes/no), switching_field
193   T5 Biomol.    → molecule_type (nucleobase/metabolite), biological_role
194   T6 Sugar      → anomeric_form (α/β), polymorph, pharmaceutical_grade
195   T7 AA salt    → counterion, salt_ratio, SHG_activity (yes/no/value)
196   T8 CT crystal → donor_molecule, acceptor_molecule, degree_of_CT (p), phase_transition_T
197
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